Nikhil Kale
47

Experiment No. 4

Aim: To develop a blockchain-powered web application using Solidity programming language on Remix IDE and embedding Metamask Wallet

THEORY

1. MetaMask

MetaMask is a cryptocurrency wallet and browser extension used to interact with blockchain networks and decentralized applications. It allows users to manage accounts and approve blockchain transactions.

2. Testnet

A testnet is a blockchain network used for development and testing. It uses test cryptocurrency, which has no real monetary value. In this experiment, Sepolia Testnet is used.

3. Remix IDE

Remix IDE is a browser-based development environment used to write, compile, deploy and interact with Solidity smart contracts.

4. Solidity Smart Contract

A smart contract is a program deployed on a blockchain that automatically executes predefined instructions. Solidity is the programming language used to develop Ethereum smart contracts.

5. Connecting MetaMask with Remix IDE

MetaMask is connected to Remix IDE using the Injected Provider or Browser Extension option. This allows users to deploy smart contracts and approve transactions through their MetaMask wallet.

6. Transactions and Gas Fees

A transaction is an operation performed on the blockchain, such as transferring ETH or executing a smart contract. Gas fees are the charges paid for processing transactions on the blockchain.

7. ETH Transfer Using Smart Contract

The ETH Transfer smart contract allows users to transfer Ether from one wallet address to another using the transferETH() function. It verifies the receiver's address and transfers the specified amount.

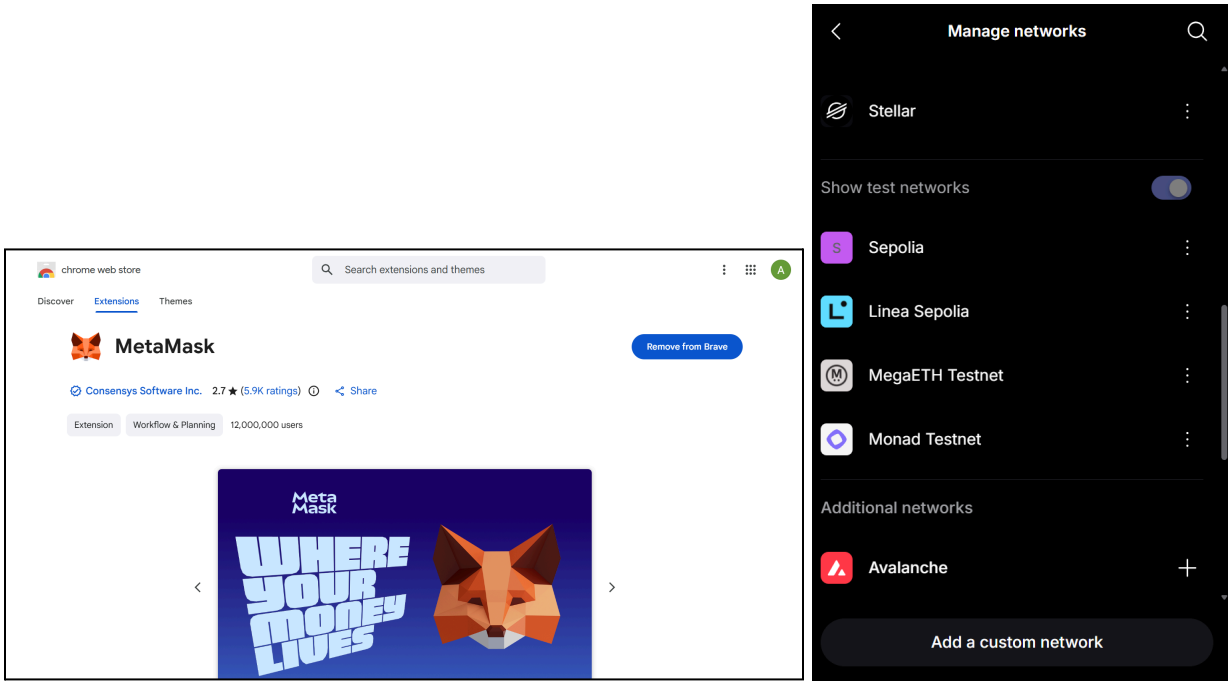
8. Blockchain Explorer

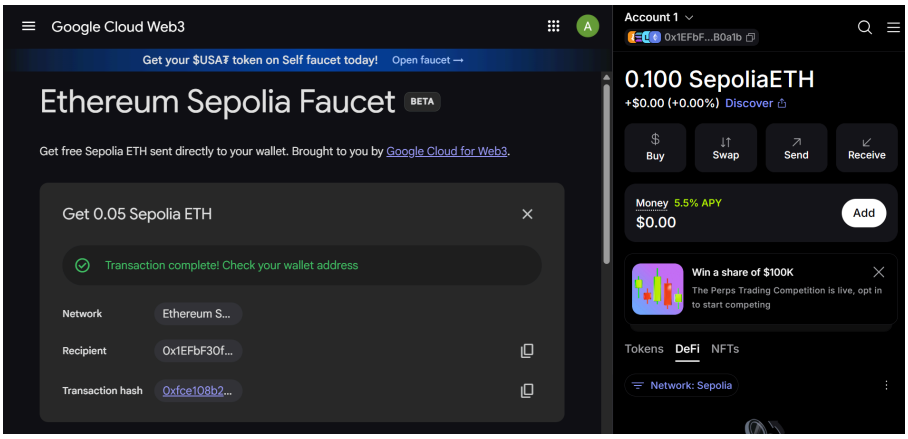
A blockchain explorer is a web-based tool used to view and verify blockchain transactions. Sepolia Etherscan is used to check transaction hashes, sender and receiver addresses, status and gas fees.

9. Interaction with Smart Contract

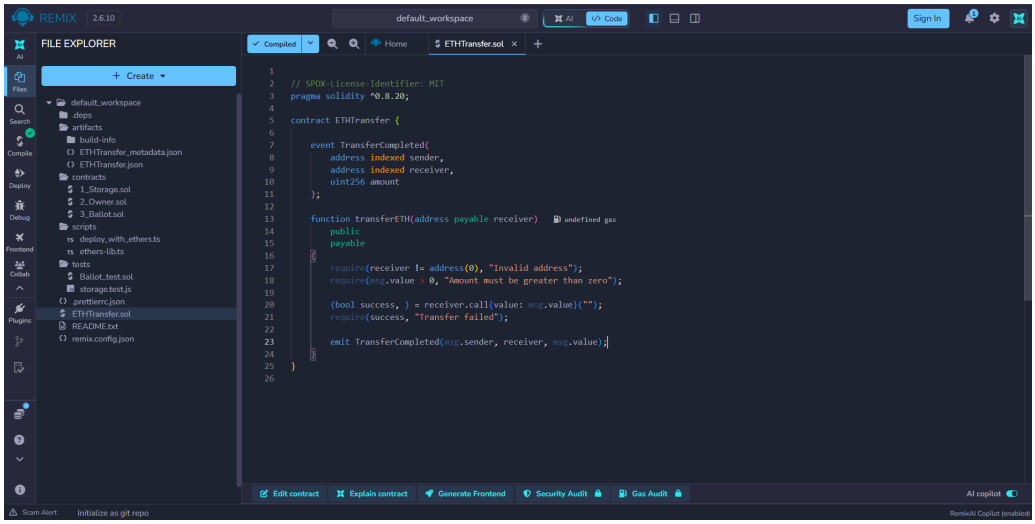
After deployment, users can interact with the smart contract through Remix IDE. MetaMask is used to approve transactions, which are then processed and recorded on the blockchain.

Output & Code:

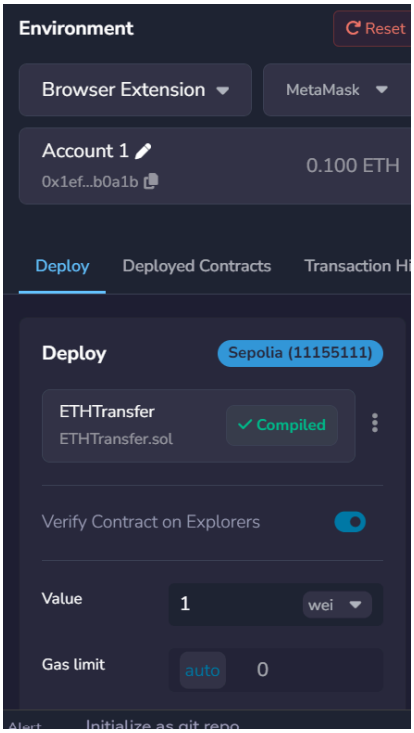




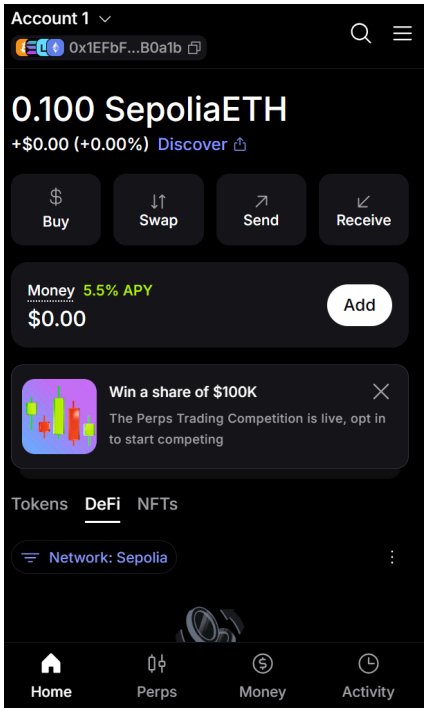
Funding MetaMask Wallet with Sepolia ETH



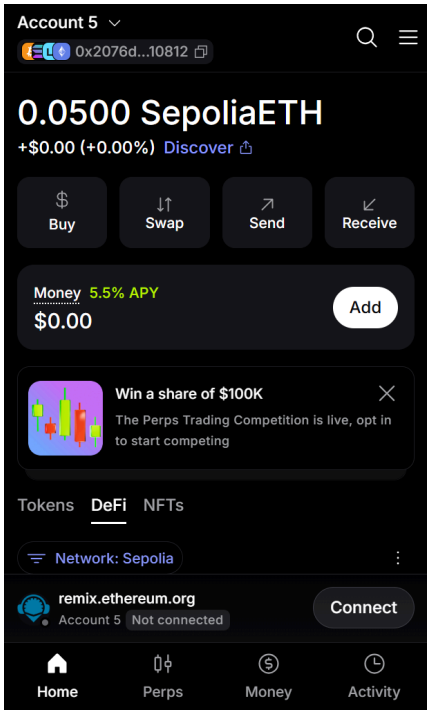
ETH Transfer Smart Contract in Remix IDE



Connecting MetaMask with Remix IDE



Account 1 (Sender Account)



Account 5 (Recipient Account)

Conclusion

Successfully developed and deployed an ETH Transfer smart contract using Solidity on Remix IDE. MetaMask was integrated to perform ETH transactions on the Sepolia Testnet. The transfer was successfully executed and verified using Sepolia Etherscan. Thus, the working of smart contracts and blockchain transactions was understood.